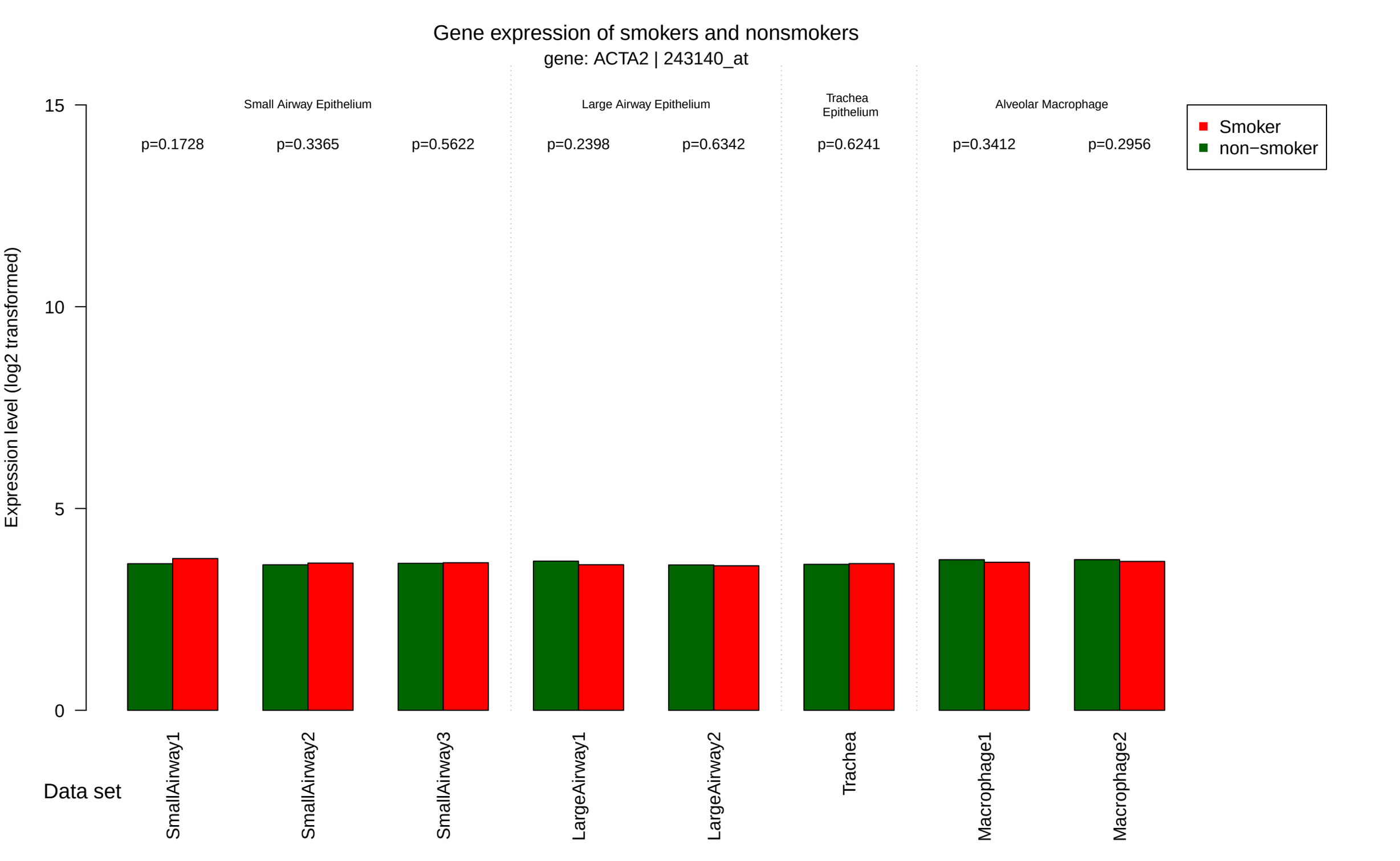
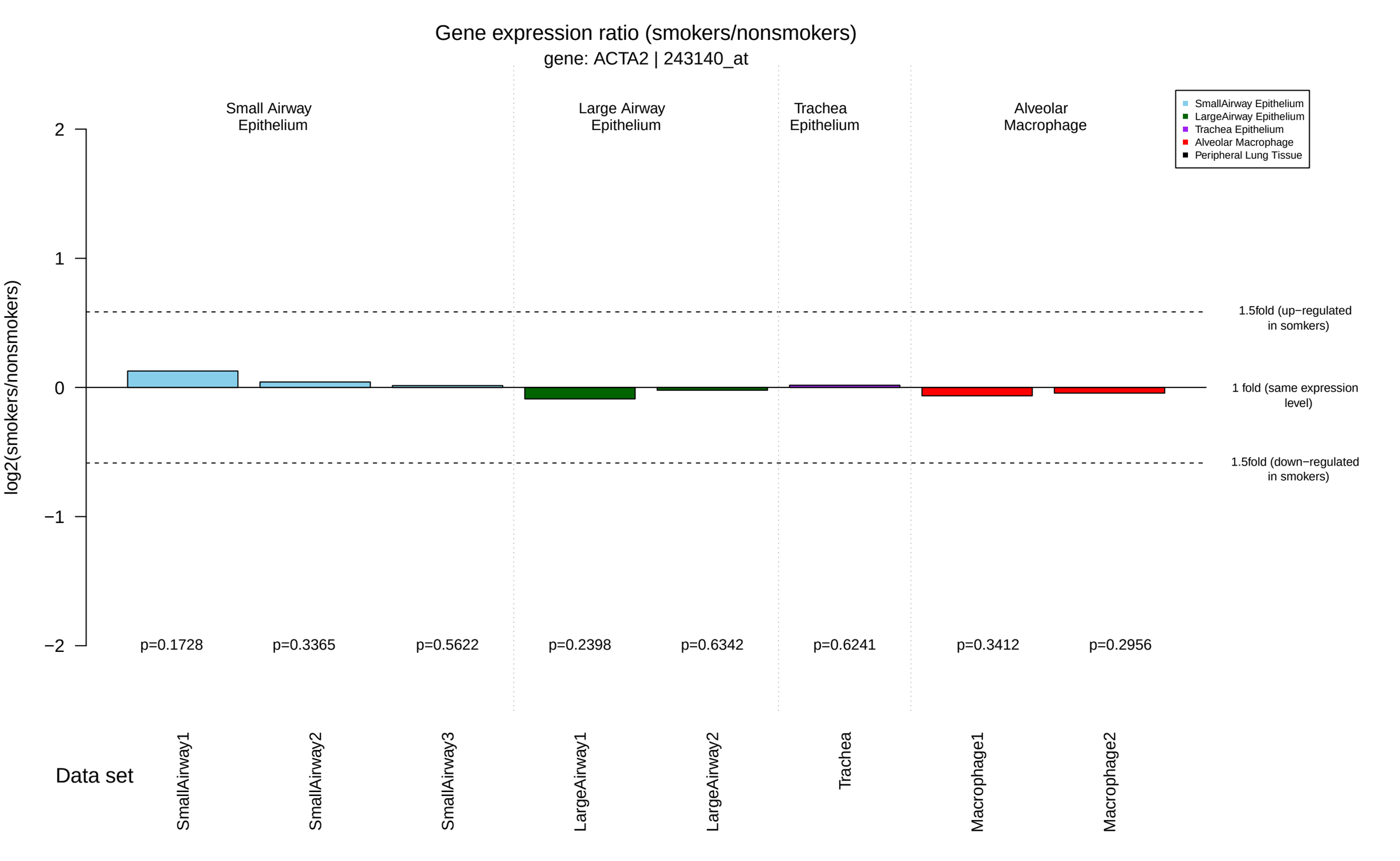
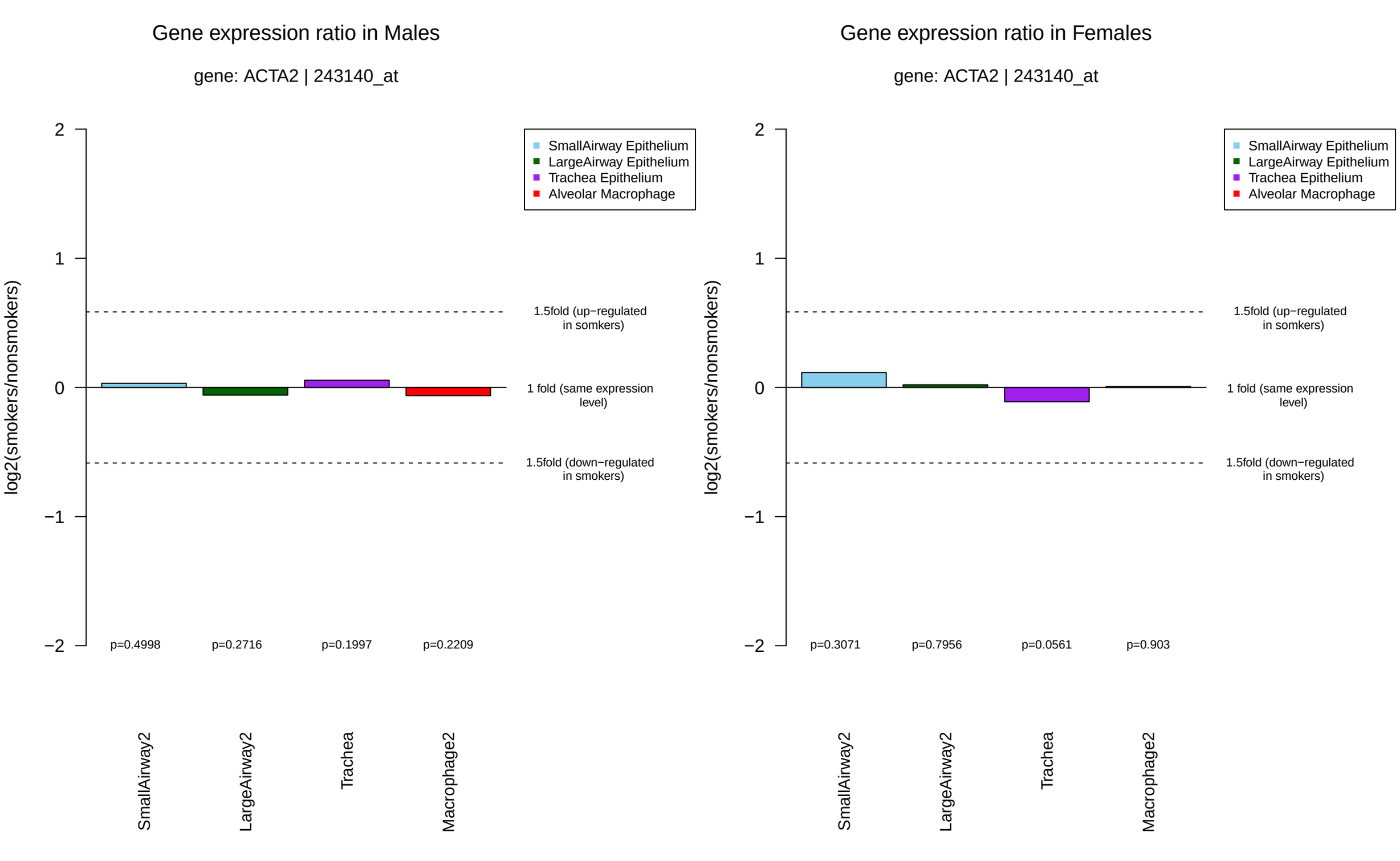
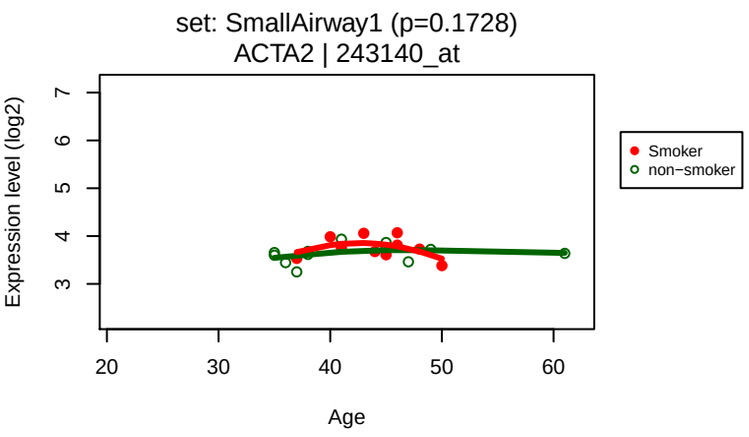
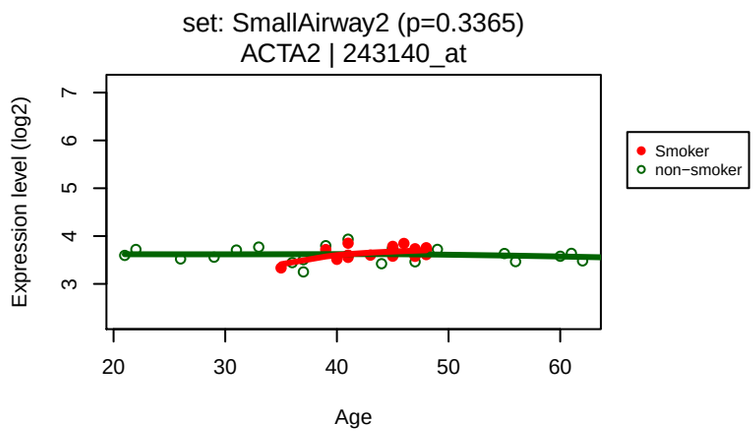
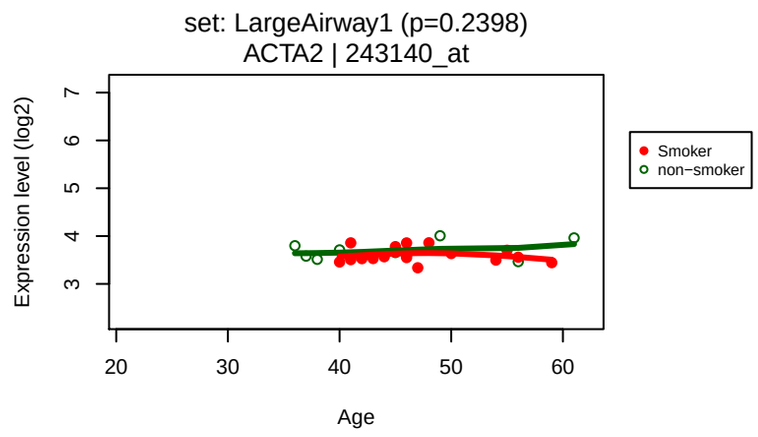

Gene expression of smokers and nonsmokers
gene: ACTA2 | 243140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1728
p=0.3365
p=0.5622
p=0.2398
p=0.6342
p=0.6241
p=0.3412
p=0.2956
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

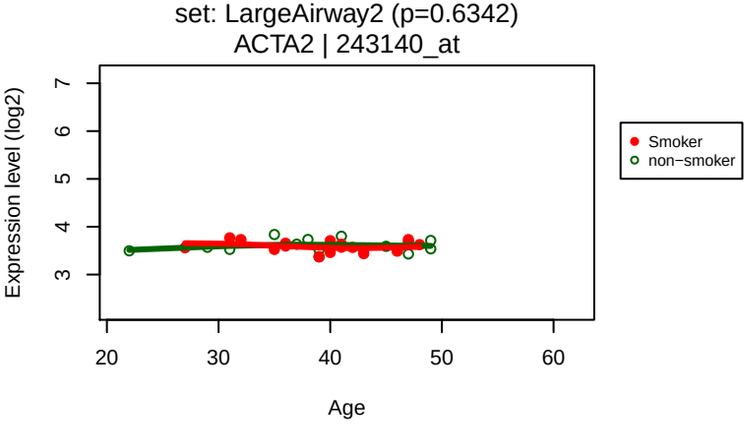

Gene expression ratio (smokers/nonsmokers)
gene: ACTA2 | 243140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1728
p=0.3365
p=0.5622
p=0.2398
p=0.6342
p=0.6241
p=0.3412
p=0.2956
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

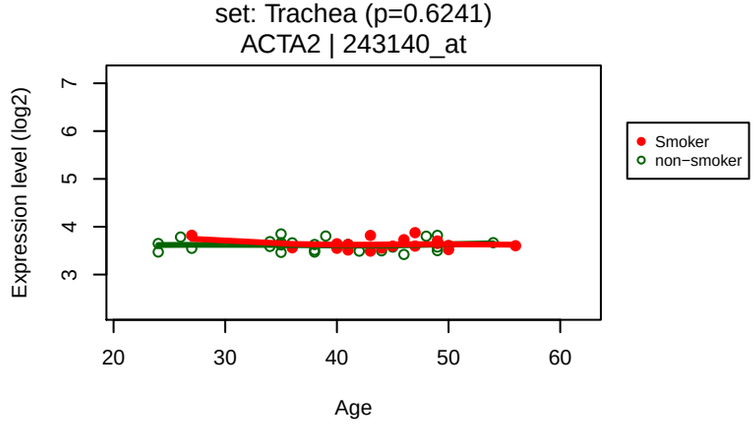

Gene expression ratio in Males
gene: ACTA2 | 243140_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4998
p=0.2716
p=0.1997
p=0.2209
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTA2 | 243140_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3071
p=0.7956
p=0.0561
p=0.903
SmallAirway2
LargeAirway2
Trachea
Macrophage2

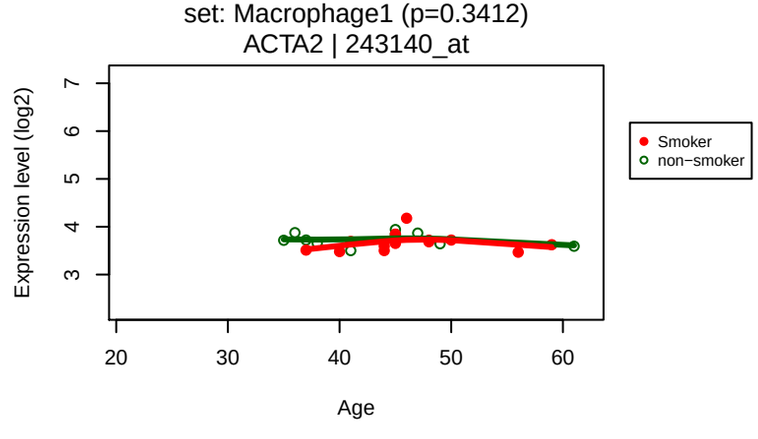

set: SmallAirway1 (p=0.1728)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker

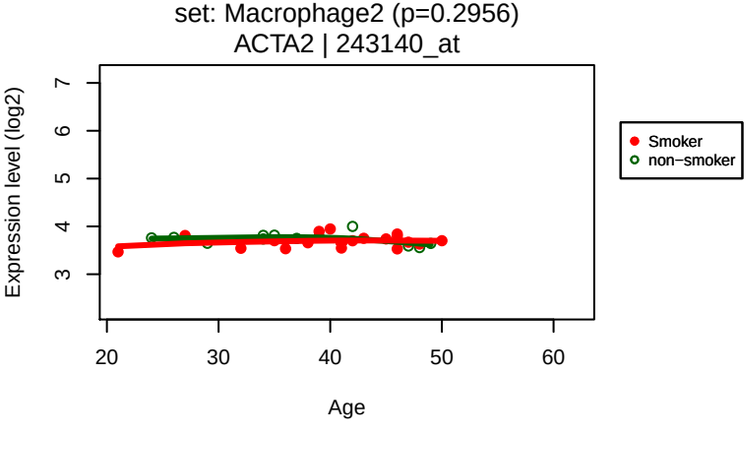

set: SmallAirway2 (p=0.3365)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker

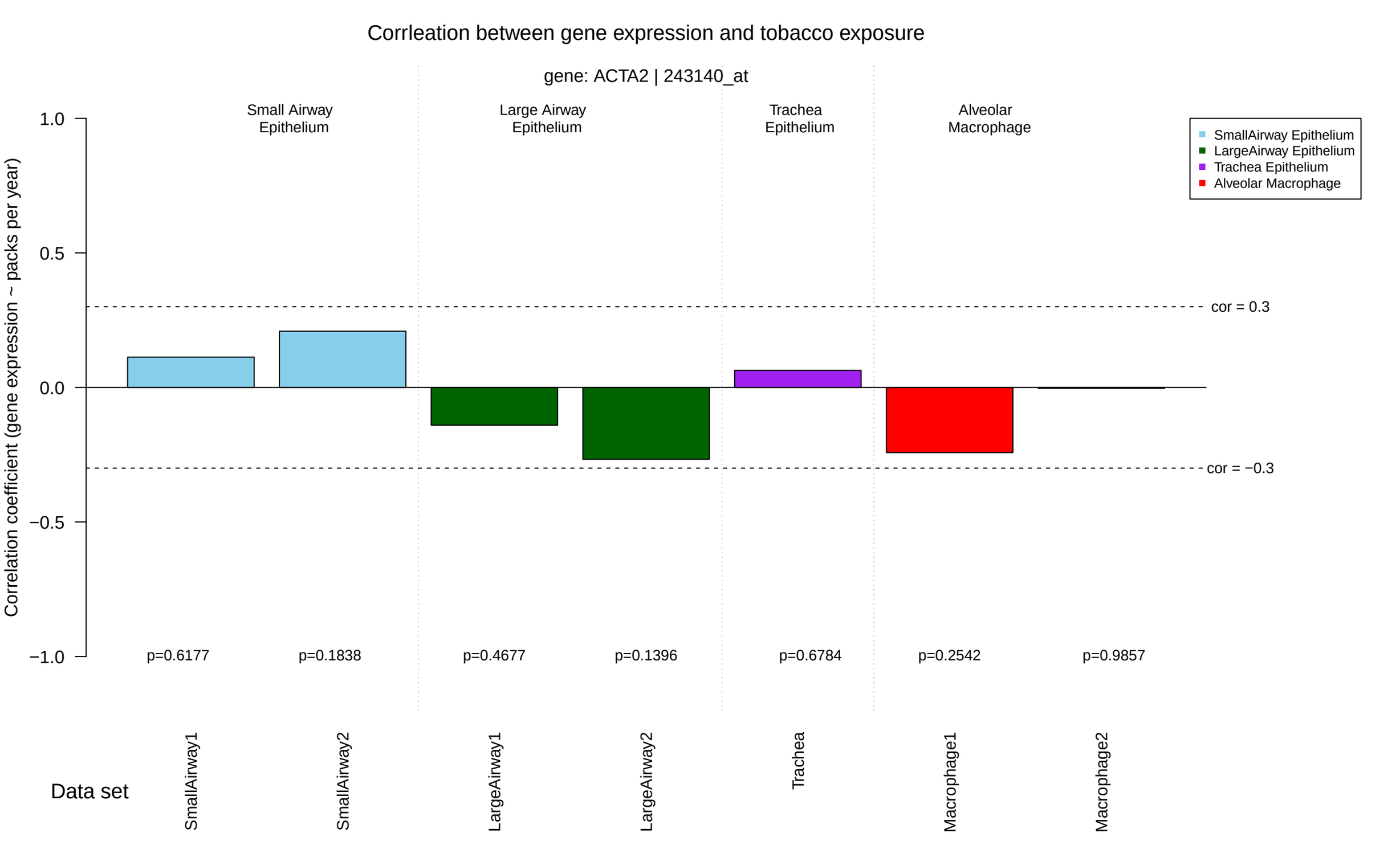

set: LargeAirway1 (p=0.2398)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6342)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6241)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3412)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2956)
ACTA2 | 243140_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTA2 | 243140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6177
p=0.1838
p=0.4677
p=0.1396
p=0.6784
p=0.2542
p=0.9857
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2